

Outbreak Intelligence: Early Warning Risk Model for Infectious Disease Vulnerability

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2026-05-08

```
library(WDI)
```

```
## Warning: package 'WDI' was built under R version 4.4.3
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.4.3
```

```
## corrplot 0.95 loaded
```

```
library(ggplot2)
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##     discard
```

```
##
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
##     col_factor
```

```
library(ggthemes)
```

```
## Warning: package 'ggthemes' was built under R version 4.4.3
```

```
library(sf)
```

```
## Warning: package 'sf' was built under R version 4.4.3
```

```
## Warning in CPL_gdal_init(): GDAL Error 1: Can't load requested DLL: C:\Program Files\GeoDa Software\
## 126: The specified module could not be found.
```

```
## Warning in CPL_gdal_init(): GDAL Error 1: Can't load requested DLL: C:\Program Files\GeoDa Software\
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```

```
## Warning in CPL_gdal_init(): GDAL Error 1: Can't load requested DLL: C:\Program Files\GeoDa Software\
## 126: The specified module could not be found.
```

```
## Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(rnaturalearth)
```

```
## Warning: package 'rnaturalearth' was built under R version 4.4.3
```

```
library(rnaturalearthdata)
```

```
## Warning: package 'rnaturalearthdata' was built under R version 4.4.3
```

```
##  
## Attaching package: 'rnaturalearthdata'  
##  
## The following object is masked from 'package:rnaturalearth':  
##  
## countries110
```

```
library(plotly)
```

```
## Warning: package 'plotly' was built under R version 4.4.3
```

```
##  
## Attaching package: 'plotly'  
##  
## The following object is masked from 'package:ggplot2':  
##  
## last_plot  
##  
## The following object is masked from 'package:stats':  
##  
## filter  
##  
## The following object is masked from 'package:graphics':  
##  
## layout
```

```
theme_set(theme_minimal(base_size = 13))
```

Load data

```
raw_data <- WDI(  
  country = "all",  
  indicator = c(  
    health_exp = "SH.XPD.CHEX.PC.CD",  
    life_exp   = "SP.DYN.LE00.IN",  
    beds       = "SH.MED.BEDS.ZS",  
    gdp        = "NY.GDP.PCAP.CD",  
    urban      = "SP.URB.TOTL.IN.ZS"  
  ),  
  start = 2018,  
  end = 2022,  
  extra = TRUE  
)
```

Clean data

```
df <- raw_data %>%
  select(country, year, health_exp, life_exp, beds, gdp, urban) %>%
  drop_na()
```

Reduce noise in Country level (Aggregate)

```
df_country <- df %>%
  group_by(country) %>%
  summarise(
    health_exp = mean(health_exp, na.rm = TRUE),
    life_exp   = mean(life_exp, na.rm = TRUE),
    beds       = mean(beds, na.rm = TRUE),
    gdp        = mean(gdp, na.rm = TRUE),
    urban      = mean(urban, na.rm = TRUE)
  )
```

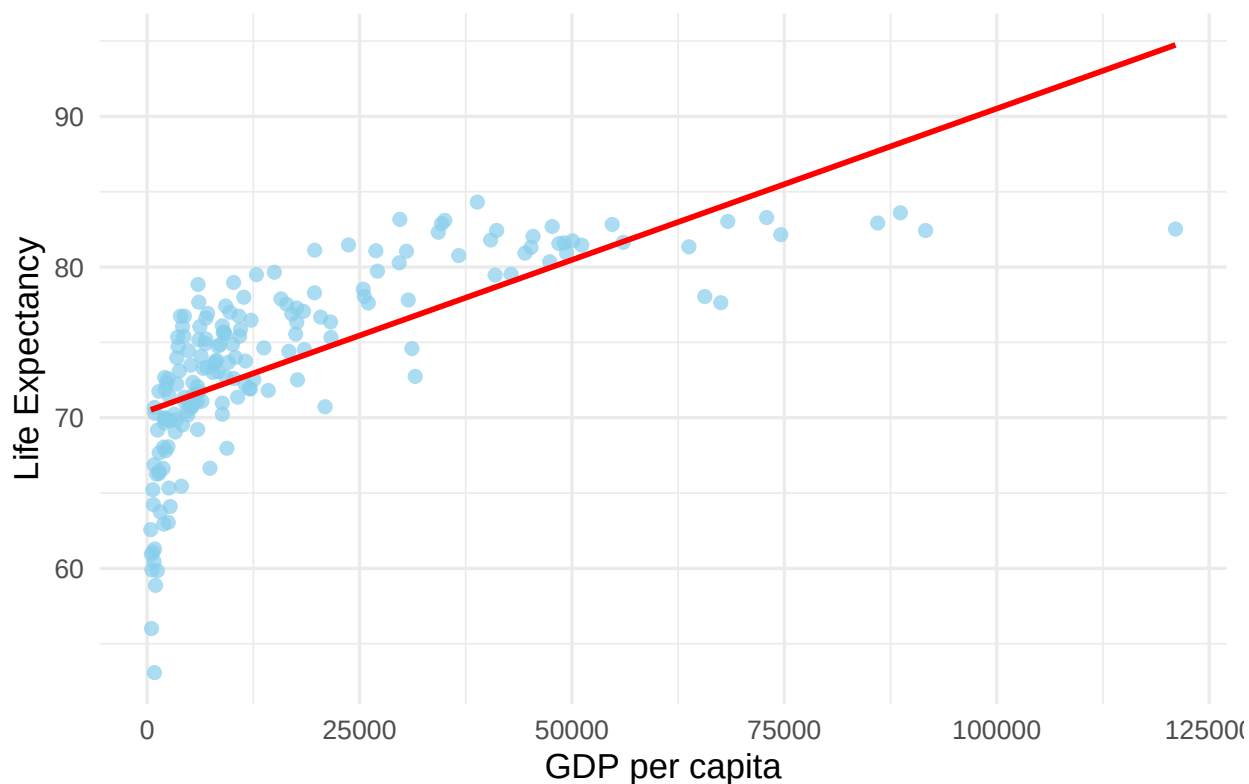
```
summary(df_country)
```

```
##      country      health_exp      life_exp      beds
## Length:181      Min.   :   12.89      Min.   :53.08      Min.   : 0.150
## Class :character 1st Qu.:  183.63      1st Qu.:70.68      1st Qu.: 1.338
## Mode  :character Median :   498.89      Median :74.00      Median : 2.636
##              Mean   : 1414.83      Mean   :73.81      Mean   : 3.039
##              3rd Qu.: 1433.75      3rd Qu.:77.89      3rd Qu.: 4.260
##              Max.   :11409.09      Max.   :84.31      Max.   :12.706
##      gdp      urban
## Min.   :   442.5      Min.   : 13.93
## 1st Qu.:  3505.4      1st Qu.: 50.97
## Median :   8366.5      Median : 65.25
## Mean   : 16822.5      Mean   : 62.64
## 3rd Qu.: 20945.7      3rd Qu.: 78.64
## Max.   :121033.8      Max.   :100.00
```

```
ggplot(df_country, aes(x = gdp, y = life_exp)) +
  geom_point(color = "skyblue", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(
    title = "GDP vs Life Expectancy",
    x = "GDP per capita",
    y = "Life Expectancy"
  ) +
  theme_minimal(base_size = 13)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

GDP vs Life Expectancy



Risk index

```
df_country <- df_country %>%  
  mutate(  
    risk_score =  
      scale(gdp)[,1] * -0.25 +  
      scale(health_exp)[,1] * -0.30 +  
      scale(beds)[,1] * -0.20 +  
      scale(life_exp)[,1] * -0.20 +  
      scale(urban)[,1] * 0.05  
  )
```

```
df_country <- df_country %>%  
  mutate(  
    risk_level = case_when(  
      risk_score >= quantile(risk_score, 0.75) ~ "High Risk",  
      risk_score >= quantile(risk_score, 0.40) ~ "Medium Risk",  
      TRUE ~ "Low Risk"  
    )  
  )
```

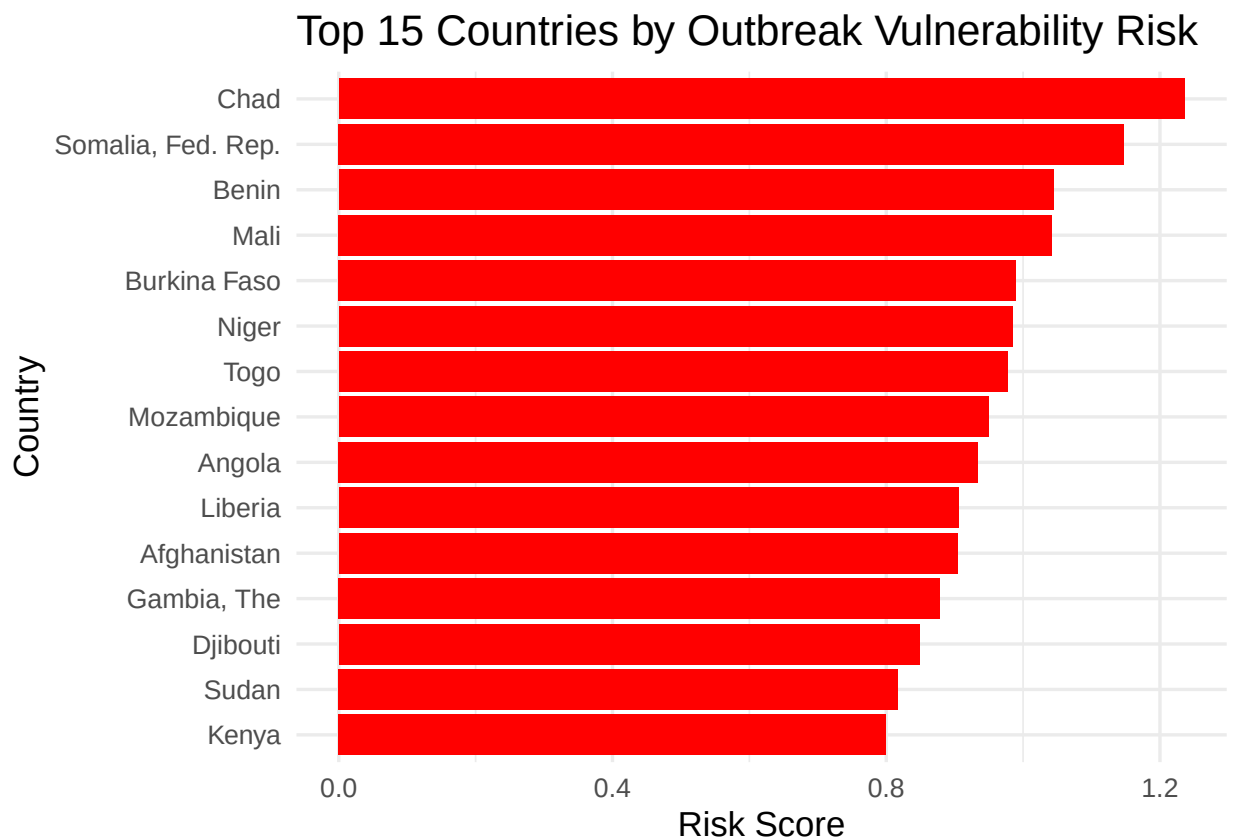
```
df_plot <- df_country %>%  
  arrange(desc(risk_score)) %>%
```

```

slice_head(n = 15)

ggplot(df_plot, aes(x = reorder(country, risk_score), y = risk_score)) +
  geom_col(fill = "red") +
  coord_flip() +
  labs(
    title = "Top 15 Countries by Outbreak Vulnerability Risk",
    x = "Country",
    y = "Risk Score"
  ) +
  theme_minimal(base_size = 13)

```



```

model <- lm(risk_score ~ gdp + health_exp + beds + life_exp + urban, data = df_country)
summary(model)

```

```

## Warning in summary.lm(model): essentially perfect fit: summary may be
## unreliable

```

```

##
## Call:
## lm(formula = risk_score ~ gdp + health_exp + beds + life_exp +
##      urban, data = df_country)
##
## Residuals:

```

```
##           Min           1Q           Median           3Q           Max
## -2.846e-15 -4.880e-17  5.600e-19  4.860e-17  1.658e-15
##
## Coefficients:
##           Estimate Std. Error   t value Pr(>|t|)
## (Intercept)  2.953e+00  4.009e-16  7.367e+15  <2e-16 ***
## gdp          -1.197e-05  2.675e-21 -4.474e+15  <2e-16 ***
## health_exp   -1.400e-04  2.402e-20 -5.829e+15  <2e-16 ***
## beds         -9.117e-02  1.166e-17 -7.818e+15  <2e-16 ***
## life_exp     -3.297e-02  6.275e-18 -5.254e+15  <2e-16 ***
## urban        2.500e-03  1.518e-18  1.646e+15  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.996e-16 on 175 degrees of freedom
## Multiple R-squared: 1, Adjusted R-squared: 1
## F-statistic: 2.28e+32 on 5 and 175 DF, p-value: < 2.2e-16
```

```
model <- lm(risk_score ~ gdp + health_exp + beds + life_exp + urban, data = df_country)

importance <- summary(model)$coefficients
```

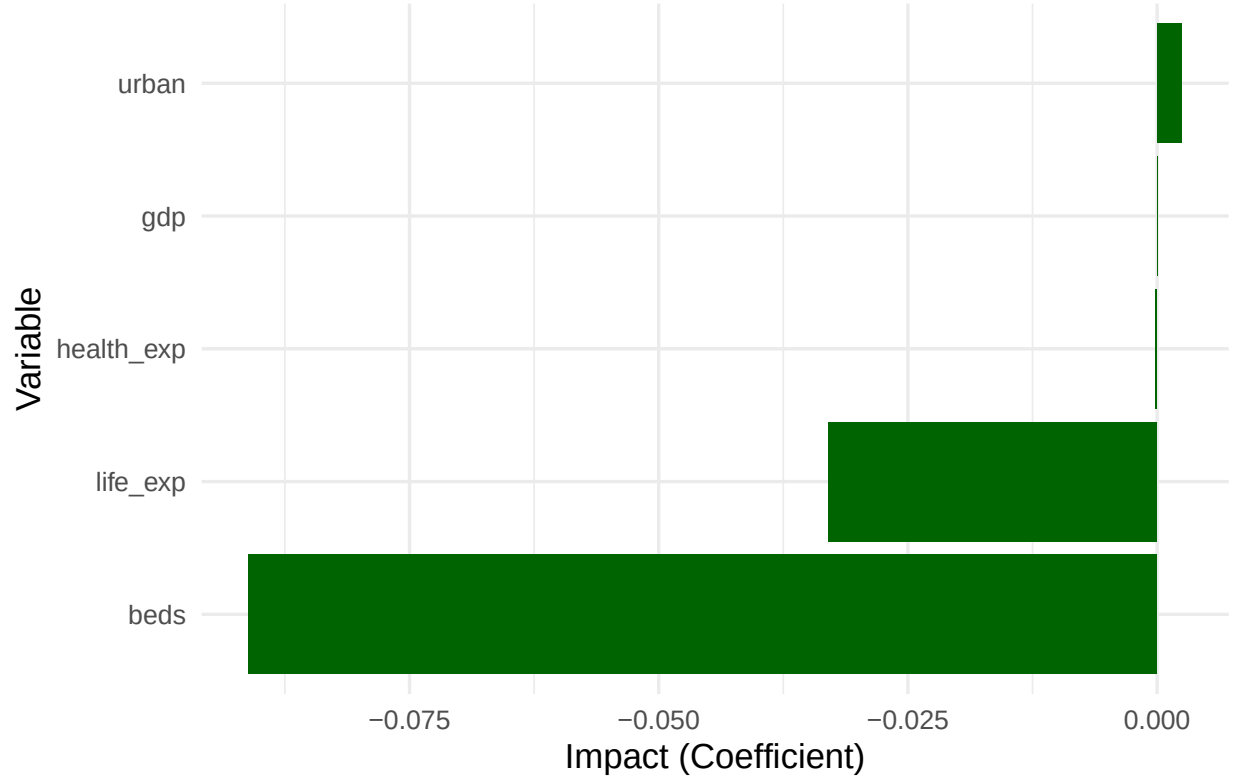
```
## Warning in summary.lm(model): essentially perfect fit: summary may be
## unreliable
```

```
importance_df <- data.frame(
  variable = rownames(importance),
  estimate = importance[,1]
)
```

```
importance_df <- importance_df %>%
  filter(variable != "(Intercept)")
```

```
ggplot(importance_df, aes(x = reorder(variable, estimate), y = estimate)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Key Drivers of Outbreak Risk",
    x = "Variable",
    y = "Impact (Coefficient)"
  ) +
  theme_minimal(base_size = 13)
```

Key Drivers of Outbreak Risk



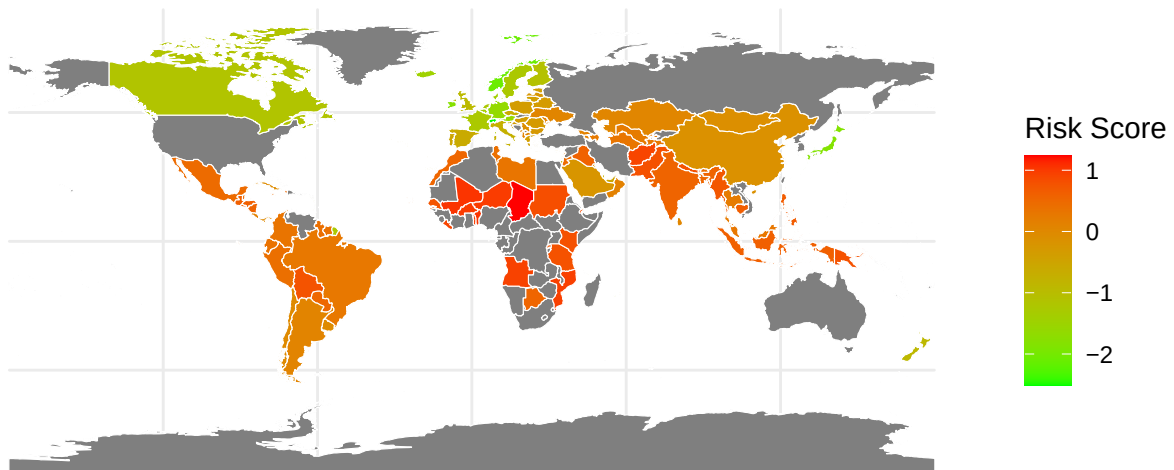
The analysis shows that healthcare expenditure and GDP per capita are the strongest protective factors

```
world <- ne_countries(scale = "medium", returnclass = "sf")
```

```
map_df <- world %>%  
  left_join(df_country, by = c("name" = "country"))
```

```
### Plotting risk map  
ggplot(map_df) +  
  geom_sf(aes(fill = risk_score), color = "white", size = 0.1) +  
  scale_fill_gradient(low = "green", high = "red") +  
  labs(  
    title = "Global Outbreak Vulnerability Risk Map",  
    fill = "Risk Score"  
  ) +  
  theme_minimal()
```


Global Outbreak Vulnerability Risk Map



```
### Standardization
df_country_scaled <- df_country %>%
  mutate(across(c(gdp, health_exp, beds, life_exp, urban), scale))
```

```
df_country <- df_country_scaled %>%
  mutate(
    risk_score =
      (-gdp) +
      (-health_exp) +
      (-beds) +
      (-life_exp) +
      (urban)
  )
```

```
### Grouping countries into risk profiles.
set.seed(123)
```

```
clusters <- kmeans(df_country_scaled %>% select(gdp, health_exp, beds, life_exp, urban),
  centers = 3)
```

```
df_country$cluster <- as.factor(clusters$cluster)
```

```
### Visualize clusters.
```

```
library(ggplot2)
library(dplyr)
library(ggmap)
```

```
## Warning: package 'ggrepel' was built under R version 4.4.3
```

```
# Create human-readable cluster labels
df_country <- df_country %>%
  mutate(
    cluster_label = case_when(
      cluster == 1 ~ "High Income / Low Risk",
      cluster == 2 ~ "Transitional Health Systems",
      cluster == 3 ~ "High Vulnerability"
    )
  )

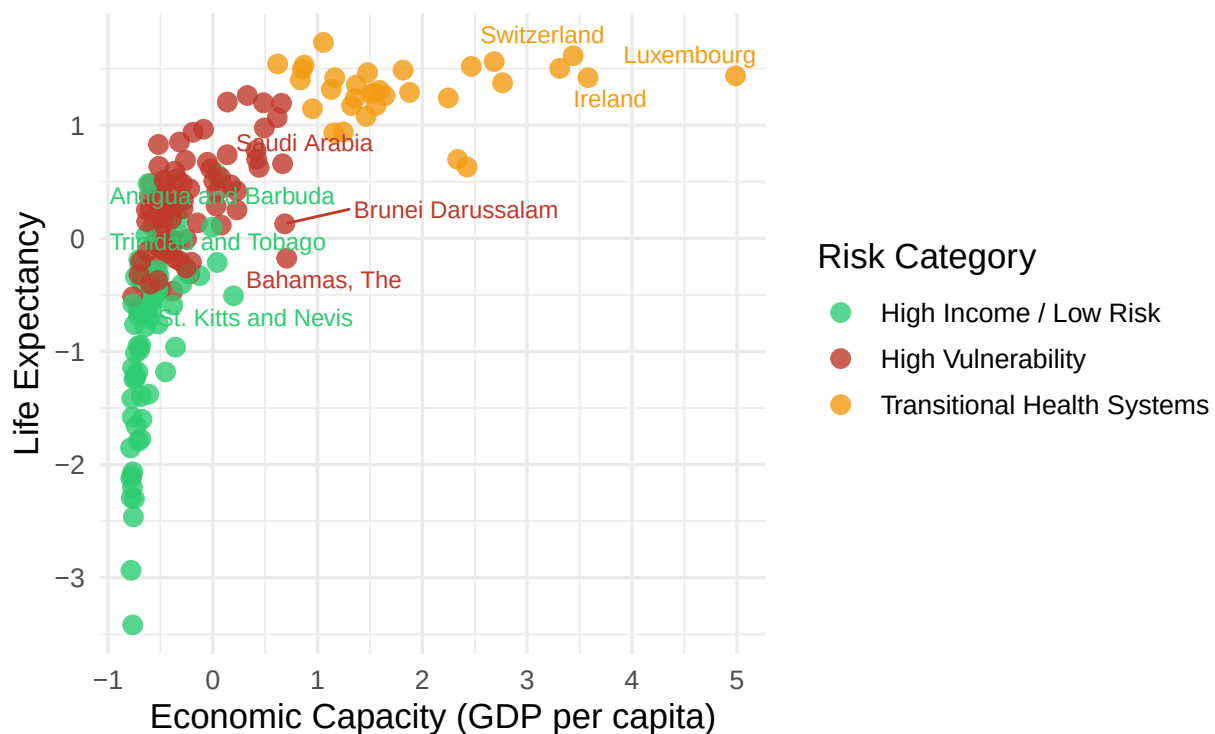
# Final visualization
ggplot(df_country, aes(x = gdp, y = life_exp, color = cluster_label)) +

  # points
  geom_point(size = 3, alpha = 0.8) +

  # optional country labels (comment out if too busy)
  geom_text_repel(
    data = df_country %>%
      group_by(cluster_label) %>%
      slice_max(gdp, n = 3),
    aes(label = country),
    size = 3,
    max.overlaps = Inf,
    show.legend = FALSE
  ) +
  scale_color_manual(values = c(
    "High Income / Low Risk" = "#2ECC71",
    "Transitional Health Systems" = "#F39C12",
    "High Vulnerability" = "#C0392B"
  )) +
  labs(
    title = "Country Health System Risk Clusters",
    subtitle = "Clusters based on economic capacity, healthcare investment, and health outcomes",
    x = "Economic Capacity (GDP per capita)",
    y = "Life Expectancy",
    color = "Risk Category"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "right"
  )
)
```

Country Health System Risk Clusters

Clusters based on economic capacity, healthcare investment, and health outcomes



This study developed a composite outbreak vulnerability index using global health and economic indicators. Results show clear stratification of countries into distinct risk groups driven primarily by healthcare investment, GDP per capita, and system capacity indicators such as hospital bed availability.

Spatial mapping reveals strong geographic clustering of high-risk countries in lower-income regions, while clustering analysis confirms structural separation between high-capacity and low-capacity health systems.

These findings suggest that outbreak preparedness is strongly determined by long-term system investment rather than short-term variability.

```
# Ensure labels exist
df_country <- df_country %>%
  mutate(
    cluster_label = case_when(
      cluster == 1 ~ "High Income / Low Risk",
      cluster == 2 ~ "Transitional Health Systems",
      cluster == 3 ~ "High Vulnerability"
    )
  )

# Build ggplot base
p <- ggplot(df_country, aes(
  x = gdp,
  y = life_exp,
  color = cluster_label,
  text = paste0(
    "Country: ", country,

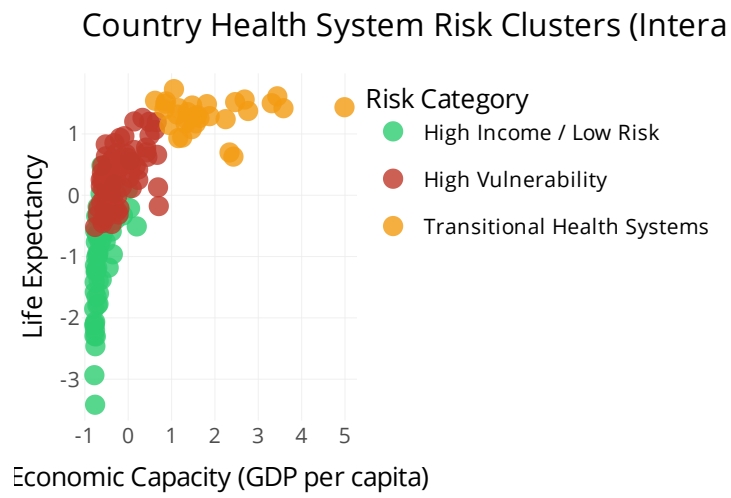
```

```

    "<br>GDP per capita: ", round(gdp, 0),
    "<br>Life Expectancy: ", round(life_exp, 1),
    "<br>Cluster: ", cluster_label
  )
)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c(
    "High Income / Low Risk" = "#2ECC71",
    "Transitional Health Systems" = "#F39C12",
    "High Vulnerability" = "#C0392B"
  )) +
  labs(
    title = "Country Health System Risk Clusters (Interactive)",
    x = "Economic Capacity (GDP per capita)",
    y = "Life Expectancy",
    color = "Risk Category"
  ) +
  theme_minimal(base_size = 13)

# Convert to plotly
ggplotly(p, tooltip = "text")

```



```
cluster_summary <- df_country %>%  
  group_by(cluster_label) %>%
```

```

summarise(
  avg_gdp = mean(gdp, na.rm = TRUE),
  avg_life_exp = mean(life_exp, na.rm = TRUE),
  avg_health_exp = mean(health_exp, na.rm = TRUE),
  avg_beds = mean(beds, na.rm = TRUE),
  n = n()
)

cluster_summary

```

```

## # A tibble: 3 x 6
##   cluster_label      avg_gdp avg_life_exp avg_health_exp avg_beds      n
##   <chr>            <dbl>      <dbl>         <dbl>    <dbl> <int>
## 1 High Income / Low Risk -0.593    -0.923        -0.551   -0.624    70
## 2 High Vulnerability   -0.217      0.288        -0.263    0.320    79
## 3 Transitional Health Systems  1.83      1.31         1.85     0.576    32

```

Cluster profiles show clear stratification of health system capacity. High-income clusters demonstr

```

# =====
# 1. CLUSTER PROFILING (DESCRIBES EACH GROUP)
# =====

cluster_summary <- df_country %>%
  group_by(cluster_label) %>%
  summarise(
    avg_gdp = mean(gdp, na.rm = TRUE),
    avg_life_exp = mean(life_exp, na.rm = TRUE),
    avg_health_exp = mean(health_exp, na.rm = TRUE),
    avg_beds = mean(beds, na.rm = TRUE),
    n = n()
  )

# View cluster structure
cluster_summary

```

```

## # A tibble: 3 x 6
##   cluster_label      avg_gdp avg_life_exp avg_health_exp avg_beds      n
##   <chr>            <dbl>      <dbl>         <dbl>    <dbl> <int>
## 1 High Income / Low Risk -0.593    -0.923        -0.551   -0.624    70
## 2 High Vulnerability   -0.217      0.288        -0.263    0.320    79
## 3 Transitional Health Systems  1.83      1.31         1.85     0.576    32

```

```

# =====
# 2. STATISTICAL TEST: ARE CLUSTERS REALLY DIFFERENT?
# =====

# One-way ANOVA to test if life expectancy differs across clusters
anova_model <- aov(life_exp ~ cluster_label, data = df_country)

summary(anova_model)

```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## cluster_label  2 120.82   60.41   181.7 <2e-16 ***
## Residuals    178   59.18    0.33
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Post-hoc test (which clusters differ from each other)
TukeyHSD(anova_model)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = life_exp ~ cluster_label, data = df_country)
##
## $cluster_label
##
##           diff          lwr          upr
## High Vulnerability-High Income / Low Risk 1.210527 0.9868286 1.434226
## Transitional Health Systems-High Income / Low Risk 2.229872 1.9390618 2.520681
## Transitional Health Systems-High Vulnerability 1.019344 0.7337787 1.304910
##
##           p adj
## High Vulnerability-High Income / Low Risk 0
## Transitional Health Systems-High Income / Low Risk 0
## Transitional Health Systems-High Vulnerability 0
```

```
# =====
# 3. STANDARDIZED MODEL (FAIR COMPARISON OF DRIVERS)
# =====

# Standardize variables so coefficients are comparable
df_std <- df_country %>%
  mutate(across(c(gdp, health_exp, beds, life_exp, urban, risk_score), scale))

# Regression: what drives outbreak risk?
model_std <- lm(risk_score ~ gdp + health_exp + beds + life_exp + urban, data = df_std)

summary(model_std)
```

```
## Warning in summary.lm(model_std): essentially perfect fit: summary may be
## unreliable
```

```
##
## Call:
## lm(formula = risk_score ~ gdp + health_exp + beds + life_exp +
##     urban, data = df_std)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.712e-15 -1.159e-16  1.030e-17  1.062e-16  4.500e-15
##
## Coefficients:
##              Estimate Std. Error   t value Pr(>|t|)
## (Intercept) -1.409e-16  3.376e-17 -4.174e+00  4.7e-05 ***
```

```
## gdp          -3.632e-01  8.469e-17 -4.289e+15 < 2e-16 ***
## health_exp  -3.632e-01  7.801e-17 -4.656e+15 < 2e-16 ***
## beds        -3.632e-01  3.877e-17 -9.367e+15 < 2e-16 ***
## life_exp    -3.632e-01  5.770e-17 -6.295e+15 < 2e-16 ***
## urban       3.632e-01  4.604e-17  7.889e+15 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.541e-16 on 175 degrees of freedom
## Multiple R-squared:      1, Adjusted R-squared:      1
## F-statistic: 1.745e+32 on 5 and 175 DF, p-value: < 2.2e-16
```

```
# =====
# 4. VISUALIZE DRIVERS OF RISK
# =====

coef_df <- data.frame(
  variable = names(coef(model_std)),
  estimate = coef(model_std)
) %>%
  filter(variable != "(Intercept)")

# Coefficient plot
ggplot(coef_df, aes(x = reorder(variable, estimate), y = estimate)) +
  geom_col(fill = "#34495E") +
  coord_flip() +
  labs(
    title = "Standardized Drivers of Outbreak Risk",
    x = "Variable",
    y = "Effect on Risk Score"
  ) +
  theme_minimal(base_size = 13)
```